

FORENXAI Technology Stack & Components

Comprehensive Overview of Architecture, Tools, and Artifacts

CATEGORY	TECHNOLOGY / COMPONENT	VERSION / CONFIGURATION	PURPOSE IN FORENXAI
Programming Language	Python	3.13	Backend services, PCAP analysis, ML inference, SHAP explanations, API processing
Programming Language	C#	Verify installed .NET SDK version	Desktop frontend logic
UI Markup	XAML	WPF-compatible	Desktop interface layout and styling
Programming Language	Java	1.8.0_202	Runs CICFlowMeter and jNetPcap components
Data Format	JSON	—	Stores and transfers forensic analysis results
Backend Framework	FastAPI	Verify locally	REST API connecting the desktop frontend to the Python backend
Backend Server	Uvicorn	Verify locally	Runs the FastAPI backend
Data Validation	Pydantic	Verify locally	Validates API requests and data models
Packet Analysis	Scapy	Verify locally	Reads PCAP files and extracts packet-level information
Data Processing	pandas	Verify locally	Loads CICFlowMeter CSV files and prepares feature tables
Numerical Computing	NumPy	2.4.6	Numerical arrays, preprocessing, SHAP and model-input handling
Machine Learning	scikit-learn	1.9.0	StandardScaler and LabelEncoder used by the trained model
Machine Learning	XGBoost	3.2.0	16-class network traffic classifier
Explainable AI	SHAP	0.51.0	Generates TreeSHAP explanations for model predictions
Model Serialization	joblib	1.5.3	Loads saved model and preprocessing artifacts

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Python Standard Library	hashlib	Built-in	SHA-256 hashing of evidence and model artifacts
Python Standard Library	pathlib	Built-in	File and directory path management
Python Standard Library	json	Built-in	Creates and reads analysis.json and manifest data
Python Standard Library	subprocess	Built-in	Launches CICFlowMeter through Maven
Python Standard Library	shutil	Built-in	Copies and stages PCAP and CSV files
Python Standard Library	os	Built-in	Environment variables and atomic file replacement
Python Standard Library	re	Built-in	Validates case ID formats
Model Artifact	XGBoost.pkl	—	Stores the trained 16-class XGBoost classifier
Model Artifact	scaler.pkl	—	Standardizes the 74 model input features
Model Artifact	label_encoder.pkl	—	Converts encoded predictions into attack-class labels
Model Artifact	features.pkl	—	Stores the exact 74-feature input order
Model Artifact	manifest.json	—	Stores versions, hashes, class labels, feature order, and artifact metadata
Model Artifact	shap_global.json	—	Stores global SHAP-related information
Flow Feature Extraction	CICFlowMeter	SimpleFlowMeterV4 0.0.4-SNAPSHOT	Converts PCAP traffic into machine-learning-compatible flow features
Native Packet Library	jNetPcap	1.4.1 / 1.4.r1425 bundle	Java packet capture and PCAP processing dependency
Packet Driver	Npcap	Installed, 64-bit	Provides Windows packet-capture compatibility for jNetPcap
Build Tool	Apache Maven	3.9.16	Executes CICFlowMeter and manages Java dependencies
Java Compiler	javac	1.8.0_202	Compiles Java components

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System Architecture	Windows architecture	64-bit / amd64	Development and runtime platform
Frontend Framework	WPF	.NET-based	Windows desktop graphical user interface
Frontend Runtime	.NET	Verify locally	Runs and builds the WPF application
Frontend Communication	HTTP Client	.NET built-in	Sends requests to the FastAPI backend
Frontend Data Handling	JSON serialization/deserialization	.NET built-in	Processes backend request and response data
Development Tool	Visual Studio Code	—	Primary source-code editor
Development Tool	PowerShell	—	Runs backend, frontend, Maven, and development commands
Development Environment	Python virtual environment (.venv)	—	Isolates Python dependencies
Version Control	Git / GitHub	—	Source-code versioning and project collaboration
Model Configuration	Input features	74	Exact CICFlowMeter-derived features used by the classifier
Model Configuration	Output classes	16	Number of traffic classes predicted by the XGBoost model